Last Time

- Finishing Effect Sizes
- Power
  - Factors affecting power
  - Power Computation
- Examples
  - The Noncentrality Parameter
- Determining Sample Size

Today

- Power
  - The Noncentrality Parameter
- Determining Sample Size
- Nonparametric Procedures
  - Bootstrapping
  - Wilcoxon Rank-Sum Test

H_0 is False: The Noncentrality Parameter

$$\begin{aligned} H_0: &\quad \mu = \mu_0 \\ H_1: &\quad \mu = \mu_1 \end{aligned}$$

Consider the test statistic $Z = \dfrac{\overline{X} - \mu_0}{\sigma/\sqrt{N}}$ when $\overline{X} \sim \mathcal{N}(\mu_1, \sigma^2)$.

Noncentrality Parameter

$$
\begin{aligned}
E(Z) &= E\left(\frac{\overline{X}-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= E\left(\frac{\overline{X}-\mu_1+\mu_1-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= E\left(\frac{\overline{X}-\mu_1}{\sigma/\sqrt{N}}\right)+E\left(\frac{\mu_1-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= 0+\frac{\mu_1-\mu_0}{\sigma/\sqrt{N}} \\
&= \frac{d}{1/\sqrt{N}} = d\sqrt{N}
\end{aligned}
$$

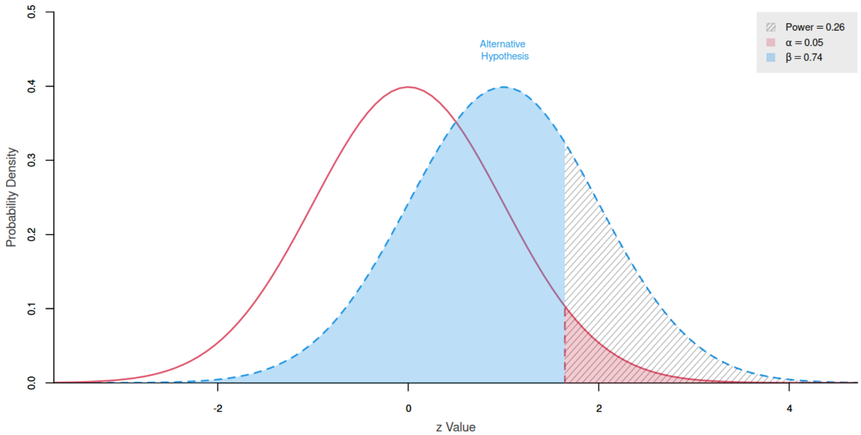
The noncentrality parameter $\delta = d\sqrt{N}$ is the amount by which the test statistic is shifted away from 0 when the null hypothesis is false.

The shape of the t distribution skews when $\delta \neq 0$ and power is no longer easy to compute by hand.

Using R (z-test)

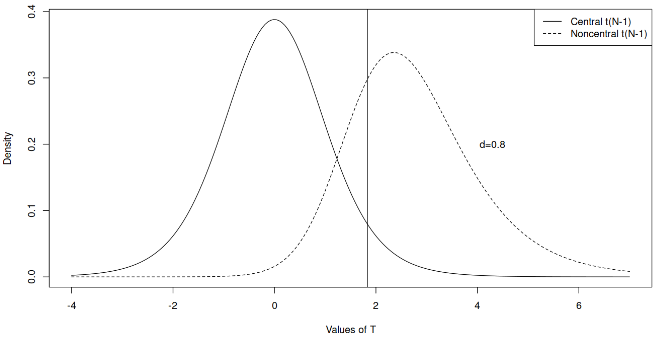
```
> library('pwrss')
> N <- 100
> d <- 0.10
> ncp <- d*sqrt(N)
> power.z.test(ncp=ncp,alpha=.05,alternative='one.sided')
+---------------------------------------------------+
|                 POWER CALCULATION                 |
+---------------------------------------------------+

Generic Z-Test

---------------------------------------------------
Hypotheses
---------------------------------------------------
  H0 (Null Claim) : mean <= null.mean
  H1 (Alt. Claim) : mean > null.mean

---------------------------------------------------
Results
---------------------------------------------------
  Type 1 Error (alpha) = 0.050
  Type 2 Error (beta)  = 0.740
  Statistical Power    = 0.26  <<
```

Using R (z-test)



One-Sample t-Test

Problem: the t-distribution shape changes when it is shifted away from 0.
The noncentrality parameter is the same as for the z-test: $d\sqrt{N}$



Katz et al. (1990)

How well do people answer SAT reading comprehension questions when they haven't read the passage to which the questions refer?

Group 1 Read Passages	Group 2 Don't Read Passages
$\overline{X_1} = 69.6$	$\overline{X_2} = 46.6$
$s_1 = 10.6$	$s_2 = 6.8$
$N_1 = 17$	$N_2 = 28$

Effect Size and Noncentrality Parameter

- Effect Size

$$\begin{aligned} s_P^2 &= \frac{(17-1)10.6^2 + (28-1)6.8^2}{17+28-2} = 70.84 \\ s_P &= \sqrt{70.84} = 8.42 \\ d &= \left| \frac{69.6 - 46.6}{8.42} \right| = 2.73 \end{aligned}$$

Effect Size and Noncentrality Parameter

- Effect Size

$$
\begin{aligned}
s_P^2 &= \frac{(17-1)10.6^2 + (28-1)6.8^2}{17+28-2} = 70.84 \\
s_P &= \sqrt{70.84} = 8.42 \\
d &= \left| \frac{69.6 - 46.6}{8.42} \right| = 2.73
\end{aligned}
$$

- Noncentrality Parameter

$$
\begin{aligned}
\delta &= \frac{d}{\sqrt{\frac{1}{N_1} + \frac{1}{N_2}}} \\
&= d\sqrt{\frac{N_1 N_2}{N_1 + N_2}}
\end{aligned}
$$

Two Independent Samples t-test

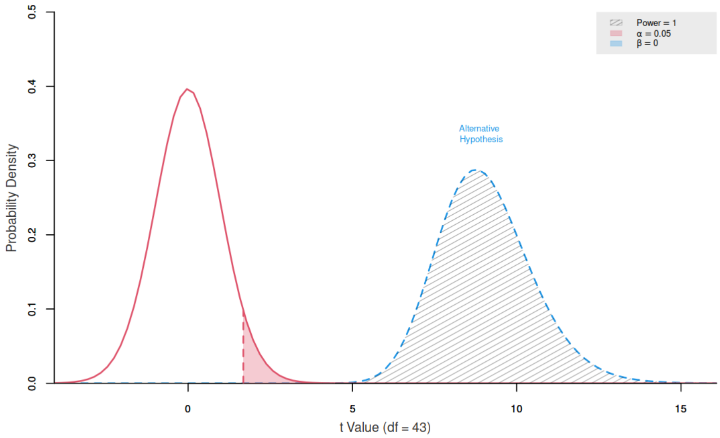
We only have the sample statistics and not the individual measurements so we have to do the t-test by hand.

```
> N.1 <- 17
> N.2 <- 28
> mean.1 <- 69.6
> mean.2 <- 46.6

> s.1 <- 10.6
> s.2 <- 6.8
> sp <- sqrt((16*10.6^2 + 27*6.8^2)/(17+28-2))
> df <- N.1 + N.2 - 2

> t <- (mean.1 - mean.2)/(sp*sqrt(1/N.1 + 1/N.2))
> 1 - pt(t,df)

1.36e-11
```

Power in R

```
> d <- abs((mean.1 - mean.2)/sp)
> d

[1] 2.732625

> power.t.welch(d=d,var.ratio=s.2^2/s.1^2,n.ratio=N.2/N.1,
                n2=17,alpha=0.05,alternative='one.sided')
+--------------------------------------------------+
|                 POWER CALCULATION                |
+--------------------------------------------------+

Welch's T-Test (Independent Samples)

---------------------------------------------------
Hypotheses
---------------------------------------------------
  H0 (Null Claim) : d - null.d <= 0
  H1 (Alt. Claim) : d - null.d > 0

---------------------------------------------------
Results
---------------------------------------------------
  Sample Size              = 28 and 17
  Type 1 Error (alpha)     = 0.050
  Type 2 Error (beta)      = 0.000
  Statistical Power        = 1  <<
```

Power in R



Power = 1
α = 0.05
β = 0
Alternative
Hypothesis
Probability Density
0.0
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t Value (df = 43)

Another Example

Scores on the SAT subtests are normally distributed with mean $\mu = 500$ and standard deviation $\sigma = 100$. Sylvan Learning Systems claimed that their study techniques resulted in a mean increase of 100 points on math subtest. If this were true, how many Sylvan students would you have to test to reject $H_0: \ \mu = 500$ at an $\alpha = 0.01$ level 95% of the time?

Problem

- Hypotheses:

$$\begin{aligned} H_0 &: \quad \mu = 500 \\ H_1 &: \quad \mu = 600 \end{aligned}$$

Problem

- Hypotheses:

$$H_0: \quad \mu = 500$$
$$H_1: \quad \mu = 600$$

- The critical Z value for the test is 2.33
- The $\bar{X}$ with Z-score 2.33 satisfies

$$2.33 = Z_{\overline{X}} = \frac{\overline{X} - \mu_0}{\sigma_{\overline{X}}}$$

Sample Size Problem

- The Z score of this $\overline{X}$ *with respect to the alternative hypothesis* is

$$Z_{\overline{X}} = -1.645 \quad = \quad \frac{\overline{X} - 600}{100/\sqrt{N}}$$

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- Solve for N:

$$500 + 2.33(100/\sqrt{N}) \quad = \quad 600 - 1.645(100/\sqrt{N})$$

Sample Size Problem

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$$\begin{aligned} Z_{\overline{X}} = -1.645 &= \frac{\overline{X} - 600}{100/\sqrt{N}} \\ \Rightarrow \overline{X} &= 600 - 1.645(100/\sqrt{N}) \end{aligned}$$

- Solve for N:

$$\begin{aligned} 500 + 2.33(100/\sqrt{N}) &= 600 - 1.645(100/\sqrt{N}) \\ 233/\sqrt{N} + 164.5/\sqrt{N} &= 600 - 500 \end{aligned}$$

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- Solve for N:

$$\begin{aligned} 500 + 2.33(100/\sqrt{N}) &= 600 - 1.645(100/\sqrt{N}) \\ 233/\sqrt{N} + 164.5/\sqrt{N} &= 600 - 500 \\ 397.5/\sqrt{N} &= 100 \end{aligned}$$

Sample Size Problem

- The Z score of this $\overline{X}$ *with respect to the alternative hypothesis* is

$$\begin{aligned} Z_{\overline{X}} = -1.645 &= \frac{\overline{X} - 600}{100/\sqrt{N}} \\ \Rightarrow \overline{X} &= 600 - 1.645(100/\sqrt{N}) \end{aligned}$$

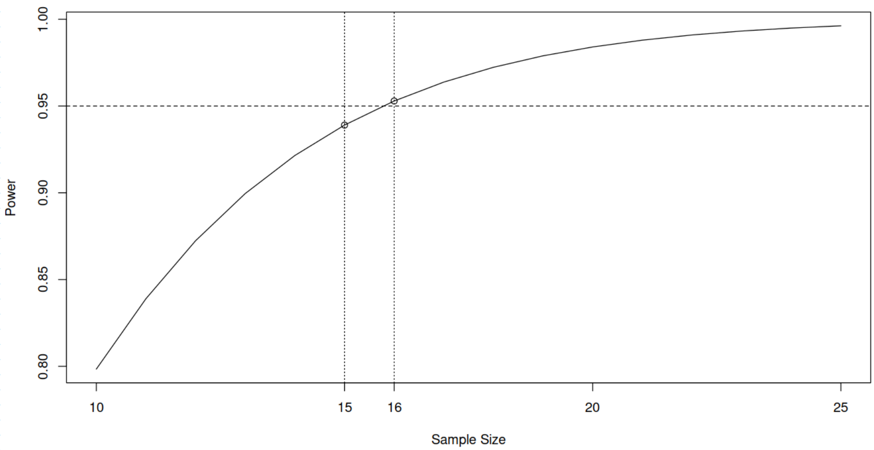
- Solve for N:

$$\begin{aligned} 500 + 2.33(100/\sqrt{N}) &= 600 - 1.645(100/\sqrt{N}) \\ 233/\sqrt{N} + 164.5/\sqrt{N} &= 600 - 500 \\ 397.5/\sqrt{N} &= 100 \\ \sqrt{N} &= 3.975 \\ N &= 3.975^2 = 15.80 \end{aligned}$$

Sample Size Determination In R

```
> library('pwrss')
> desired.power <- 0.95
> d <- 1.0
> N <- seq(10,25)
> ncp <- d * sqrt(N)
> power <- power.z.test(ncp,alpha=alpha,alternative='greater',plot=FALSE)

> first<-which(power>=desired.power)[1]
> plot(N,power,xlab="Sample Size",ylab="Power",type='l')
> points(N[(first-1):first],power[(first-1):first])
> abline(h=desired.power,lty=2)
> abline(v=c(N[first],N[first-1]),lty=3)
> axis(1,at=16,tick=TRUE)
```

Power Curve



Two-sample t-Test

```
> alpha <- 0.05
> desired.power <- 0.80
> d <- 0.5

> power <- power.t.welch(d=d,alpha=alpha,alternative='one.sided',
                         n.ratio=1,power=desired.power)
+--------------------------------------------------+
|             SAMPLE SIZE CALCULATION              |
+--------------------------------------------------+

Welch's T-Test (Independent Samples)

---------------------------------------------------
Hypotheses
---------------------------------------------------
  H0 (Null Claim) : d - null.d <= 0
  H1 (Alt. Claim) : d - null.d > 0

---------------------------------------------------
Results
---------------------------------------------------
  Sample Size            = 51 and 51  <<
  Type 1 Error (alpha)   = 0.050
  Type 2 Error (beta)    = 0.194
  Statistical Power      = 0.806
```

Power Curves and Tables

NUMBER OF OBSERVATIONS FOR *t*-TEST OF DIFFERENCE BETWEEN TWO MEANS

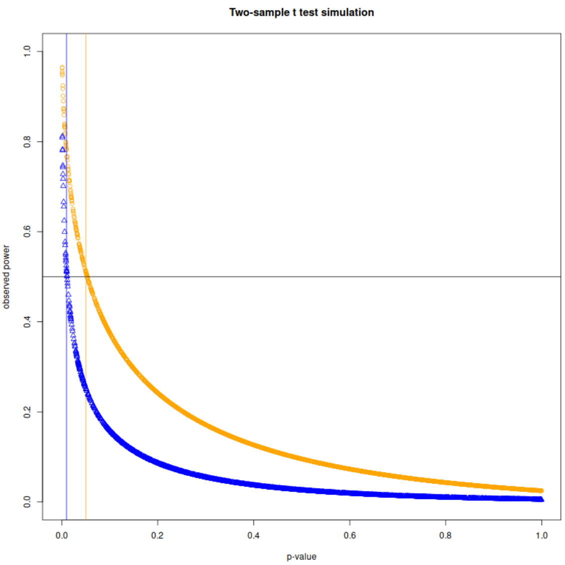
	Level of *t*-test																				
Single-sided test	$\alpha = 0.005$					$\alpha = 0.01$					$\alpha = 0.025$					$\alpha = 0.05$					
Double-sided test	$\alpha = 0.01$					$\alpha = 0.02$					$\alpha = 0.05$					$\alpha = 0.1$					
$\beta =$	0.01	0.05	0.1	0.2	0.5	0.01	0.05	0.1	0.2	0.5	0.01	0.05	0.1	0.2	0.5	0.01	0.05	0.1	0.2	0.5	
0.05																					0.05
0.10																					0.10
0.15																					0.15
0.20																				137	0.20
0.25															124					88	0.25
0.30										123					87					61	0.30
0.35					110					90					64				102	45	0.35
0.40					85					70				100	50			108	78	35	0.40
0.45				118	68				101	55			105	79	39		108	86	62	28	0.45
0.50				96	55			106	82	45		106	86	64	32		88	70	51	23	0.50
0.55			101	79	46		106	88	68	38		87	71	53	27	112	73	58	42	19	0.55
0.60		101	85	67	39		90	74	58	32	104	74	60	45	23	89	61	49	36	16	0.60
0.65		87	73	57	34	104	77	64	49	27	88	63	51	39	20	76	52	42	30	14	0.65
0.70	100	75	63	50	29	90	66	55	43	24	76	55	44	34	17	66	45	36	26	12	0.70
0.75	88	66	55	44	26	79	58	48	38	21	67	48	39	29	15	57	40	32	23	11	0.75
0.80	77	58	49	39	23	70	51	43	33	19	59	42	34	26	14	50	35	28	21	10	0.80
0.85	69	51	43	35	21	62	46	38	30	17	52	37	31	23	12	45	31	25	18	9	0.85
0.90	62	46	39	31	19	55	41	34	27	15	47	34	27	21	11	40	28	22	16	8	0.90
Value of 0.95	55	42	35	28	17	50	37	31	24	14	42	30	25	19	10	36	25	20	15	7	0.95
$\Delta = \frac{\mu_x - \mu}{\sigma}$ 1.00	50	38	32	26	15	45	33	28	22	13	38	27	23	17	9	33	23	18	14	7	1.00
1.1	42	32	27	22	13	38	28	23	19	11	32	23	19	14	8	27	19	15	12	6	1.1
1.2	36	27	23	18	11	32	24	20	16	9	27	20	16	12	7	23	16	13	10	5	1.2
1.3	31	23	20	16	10	28	21	17	14	8	23	17	14	11	6	20	14	11	9	5	1.3
1.4	27	20	17	14	9	24	18	15	12	8	20	15	12	10	6	17	12	10	8	4	1.4
1.5	24	18	15	13	8	21	16	14	11	7	18	13	11	9	5	15	11	9	7	4	1.5
1.6	21	16	14	11	7	19	14	12	10	6	16	12	10	8	5	14	10	8	6	4	1.6
1.7	19	15	13	10	7	17	13	11	9	6	14	11	9	7	4	12	9	7	6	3	1.7

Other Resources

- R pwr package
- R stats package (power.t.test,power.prop.test,power.anova.test)
  - Note that power.t.test in the pwrss package overwrites the power.t.test function in the stats package that is loaded into R by default.
- G*Power (Mac and Windows)

Post-hoc Power Calculations Are Not Informative

$$
\begin{aligned}
t(N_1 - 1 + N_2 - 1) \\
&= \frac{\overline{X}_1 - \overline{X}_2}{s_P\sqrt{\frac{1}{N_1} + \frac{1}{N_2}}} \\
&= \frac{d}{\sqrt{\frac{1}{N_1} + \frac{1}{N_2}}} \\
&= d\sqrt{\frac{N_1 N_2}{N_1 + N_2}} \\
&= \delta
\end{aligned}
$$



Nonparametric Procedures

Sometimes none of the assumptions required for the t-test hold. Sometimes we don't know what the sampling distribution of a statistic is, or we don't want to estimate unknown parameters with sample statistics.

- Nonparametric or Distribution-free
  - Resampling
    - Rank tests (ranks)
    - Bootstrapping (raw scores)
    - Randomization tests
    - Permutation tests

The Population

We don’t know the population, but for these procedures we treat the data distribution as an estimate of the population distribution.

With vs. Without Replacement

Bootstrapping: with replacement

Permution: without replacement

Bootstrapping Example

Construct a 95%confidence interval for the *median* age of GSS respondents.

A large number of times:

- Sample N observations from the data with N observations
- Compute the median for each new sample

Compute the 2.5% and 97.5% quantiles of the medians

Bootstrapping Example

```
> # Read in data
> data <- read.csv('GSS_1980.csv')
> age <- data$AGE
> N <- length(age) # sample size

> # Resample
> bignum <- 1000
> medians <- vector()
> for (i in 1:bignum)
>   medians[i] <- median(sample(age,N,replace=TRUE))

> # Compute the quantiles
>   interval.95 <- quantile(medians,c(.025,.975))

 2.5% 97.5%
   35    37
```

Two Independent Samples

Imagine each person's data in one of two boxes marked "Treatment" and "Control."
You know what the mean of the data in each box is.
Then oops! You drop the boxes and the data falls to the floor with no way of putting the observations back in the right box.

$$
\begin{aligned}
H_0 : \quad & F_X = F_Y \\
H_1 : \quad & F_X \neq F_Y
\end{aligned}
$$

Two Independent Samples

Compute the mean difference of the original treatment and control groups. Let N_X and N_Y be the sample sizes for the treatment and control groups, respectively.

A large number of times:

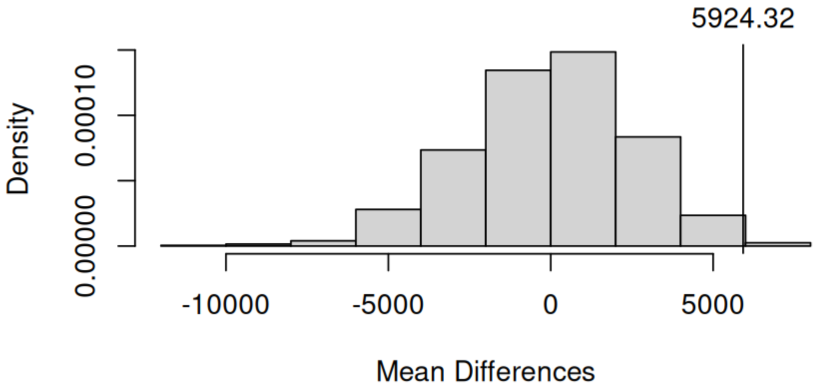
- Select N_X observations from the scrambled data for a new treatment group, put the rest in the control.
- Compute the mean difference.

What is the percentile rank of the original mean difference with respect to the resampled distribution?

Two Independent Samples

```
> happy <- data$HAPPY < 3
> money <- data$REALRINC

> # Compute the mean difference
> obs.diff <- mean(money[happy]) - mean(money[!happy])

> bignum <- 1000
> N.happy <- sum(happy)
> N.total <- length(happy)

> # Resample mean differences
> diff <- vector()
> for (i in 1:bignum) {
>   choose <- sample(N.total,N.happy,replace=FALSE)
>   diff[i] <- mean(money[choose]) - mean(money[-choose])
> }

> # Percentile rank of obs.diff
> cat("The percentile rank of ",round(obs.diff,2)," is ",mean(diff < obs.diff),
>     ", which corresponds to a 'p-value' of ",2*mean(diff >= obs.diff),".\n")

The percentile rank of  5924.32  is  0.998 , which corresponds to a 'p-value' of  0.004 .
```

Two Independent Samples



Dependent Samples

Permutation test: consider pair X_i, Y_i, e.g., pre- vs. post-treatment, where $X \sim F_X$ and $Y \sim F_Y$.

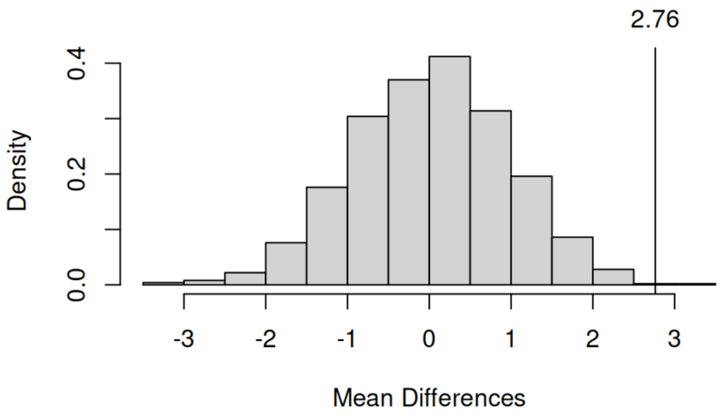
$$\begin{aligned} H_0 : \quad & F_X = F_Y \\ H_1 : \quad & F_X \neq F_Y \end{aligned}$$

If H_0 is true, it shouldn't matter whether the X score or the Y score is assigned to the pre- or post-treatment condition (permute the observations).

Permutation Test

```
> data(anorexia)

> # Compute the difference scores
> obs.diff <- mean(anorexia$Postwt - anorexia$Prewt)
> N <- length(anorexia$Prewt)

> bignum <- 1000
> mean.diffs <- vector()
> for (i in 1:bignum) {
>   # Random vector of -1,+1
>   signs <- 2*rbinom(N,1,.5) - 1
>   mean.diffs[i] <-
>     mean(signs*(anorexia$Postwt - anorexia$Prewt))
> }

> # Percentile rank of obs.diff
> cat("The percentile rank of ",round(obs.diff,2)," is ",
>     mean(mean.diffs < obs.diff),", which corresponds to a 'p-value' of ",
>     mean(mean.diffs >= obs.diff),".\n")

The percentile rank of  2.76  is  0.999 , which corresponds to a 'p-value' of  0.001 .
```

Permutation Test



Wilcoxon Rank Sum Test

$$H_0: \quad F_X = F_Y$$
$$H_1: \quad F_X \neq F_Y$$

- Consider $X = \{X_1, X_2, \ldots, X_{N_X}\}$ and $Y = \{Y_1, Y_2, \ldots, Y_{N_Y}\}$. If $\mu_X > \mu_Y$, if we rank-order all the scores in both groups from lowest to highest, the higher ranks should mostly be in group X and the lower ranks should be in group Y. If we then add those ranks, the sum of the ranks in group X should be higher than the sum of the ranks in group Y.
- If $F_X = F_Y$, then if we rank all the scores then the sum of the ranks for both groups should be about the same.
- Test statistic W_S is the sum of the ranks in the smaller group.

GSS Survey

- Under the theory driving the (ersatz) birth order effect, only children should feel more obligation toward their parents than children with siblings.
- Children who adhere to their parents' values may receive more support from their parents, resulting in greater opportunities, higher education, and ultimately higher-paying jobs.
- Only children who agree with the sentiment that younger people should be taught to respect the opinions of their elders may have higher salaries than those who disagree.

Mean Salaries:	
Agree:	\$26,141.33
Disagree:	\$26,910.00

Median Salaries:	
Agree:	\$24,547.00
Disagree:	\$15,294.50

Wilcoxon Rank Sum Test By Hand

```
> # Look at only children
> only <- data$SIBS==0

> # Subset the data
> groups <- data$YOUNGEN[only] # 1 is "agree"
> money <- data$REALRINC[only]
> N <- length(money)

> # Figure out which group has the smaller N
> smaller <- sum(groups==1) # Assume it's the agree group
> small.group <- 1
> # Check and swap to group 2 if it's smaller
> if (smaller > N/2) {
>   smaller <- N - smaller
>   small.group <- 2
> }

> # Order the ranks for all only children
> ranks <- rank(money,ties.method="average")

> # Compute the summed ranks and its p-value
> W.s <- sum(ranks[groups==small.group]) - smaller*(smaller+1)/2
> W.s

[1] 200

> 1 - pwilcox(W.s,smaller,N-smaller)

[1] 0.07936765
```

Using wilcox.test()

If sample sizes are large this can take a very long time or even crash.

```
> # Look at only children
> only <- data$SIBS==0

> # Subset the data
> groups <- data$YOUNGEN[only] # 1 is "agree"
> money <- data$REALRINC[only]

> wilcox.test(money~groups,alternative="greater",correct=FALSE,exact=FALSE)

 Wilcoxon rank sum test

data:  money by groups
W = 200, p-value = 0.07965
alternative hypothesis: true location shift is greater than 0
```

Using Resampling

```
> bignum<- 10000
> W <- vector()
> for (i in 1:bignum) W[i] <- sum(sample(N,smaller,replace=FALSE))

> shift <- smaller*(smaller+1)/2
> lower <- 0
> upper <- sum((N - smaller + 1):N) - shift

> W <- W - shift
> hist(W,freq=FALSE,main="",breaks=50,
>       xlab="Values of Ws",ylab="Probability",xlim=c(lower,upper))
> abline(v=W.s)
> text(W.s,.0122,W.s,xpd=NA)
> mean(W>=W.s)

[1] 0.0882
```

Resampling Distribution of W_s